

Problem-Solution Fit Canvas

Date	Team ID	Project Name	Maximum Marks
29 September 2026	6ab7959e4c4a9ef0de66bbbee	AI FitTrack API	5 Marks

Source-grounded canvas: the supplied API configuration explicitly supports a backend/API service, JWT authentication, MongoDB storage, and Gemini AI integration. Detailed user research, devices, frequency measurements, and offline channels are not specified in the source, so those points are kept appropriately qualified rather than invented.

Canvas Section	AI FitTrack API — Source-Based Answer
1. CUSTOMER SEGMENT(S) [CS]	Users/clients of the AI FitTrack API who interact with the fitness application or client consuming the API. The source does not specify demographic segments.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR]	Need for a backend service that can securely handle client requests, authenticate access, store application data, and provide AI-assisted functionality. Frequency of these problems is not specified in the source.
3. TRIGGERS TO ACT [TR]	A need to use the fitness application's API functionality, securely access protected operations, store/retrieve application data, or use configured AI-assisted processing. Specific real-world triggers are not specified.
4. EMOTIONS BEFORE / AFTER [EM]	Before: concern about secure access, reliable data handling, and obtaining AI-assisted functionality through the application. After: expected greater confidence from JWT-based access and centralized API/data/AI integration. These are solution-oriented expectations, not measured user research.
5. AVAILABLE SOLUTIONS PROS & CONS [AS]	Available components identified in the source: JWT-based authentication, MongoDB, and Gemini AI integration. Pros: secure access mechanism, persistent database storage, and external AI capability. Cons/limitations: the source does not document comparative alternatives or measured shortcomings.
6. CUSTOMER LIMITATIONS [CL]	Source-supported limitation: the solution depends on configured database and Gemini AI service connectivity. Specific customer budget, device, connectivity, or accessibility limitations are not specified.
7. BEHAVIOR + ITS INTENSITY [BE]	Users/clients send requests to the API, authenticated requests are processed, application data can be handled through MongoDB, and AI-assisted requests can use Gemini. Behavior intensity or usage frequency is not specified.
8. CHANNELS OF BEHAVIOR [CH]	ONLINE: API/client requests and configured external AI/database integrations. OFFLINE: no offline channel is specified in the source configuration.
9. PROBLEM ROOT / CAUSE [RC]	The source indicates a need for a backend integration layer connecting secure authentication, application data storage, and AI-assisted functionality. Specific root causes from customer interviews or research are not provided.
10. YOUR SOLUTION [SL]	AI FitTrack API: a backend API configured to receive and respond to client requests, provide JWT-based secure authentication, use MongoDB for application data storage, and integrate Gemini AI (configured model: gemini-3.6-flash) for AI-assisted functionality.

Note: The uploaded template contains the ten Problem-Solution Fit Canvas fields and specifies a maximum of 5 marks. The answers above are limited to what can be supported by the provided AI FitTrack API source configuration.